

TRABZON, Turkey

WMO: 170380

Lat: 40.995N

Lon: 39.783E

Elev: 112

StdP: 14.64

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.2	35.2	18.7	14.1	46.7	22.4	16.9	46.7	26.7	50.9	23.6	52.4	7.8	240	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	84.1	75.0	82.5	74.1	81.0	73.2	77.1	82.5	75.8	81.3	74.6	80.3	7.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	132.9	81.9	73.7	125.8	80.7	72.3	120.1	79.7	40.7	82.4	39.4	81.7	38.3	80.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	17.9	14.6	DB	30.3	88.6	2.8	3.7	28.3	91.3	26.6	93.5	25.1	95.6	23.0	98.3
			WB	27.1	78.3	2.5	1.8	25.2	79.6	23.8	80.7	22.4	81.7	20.5	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	46.8	46.5	48.8	53.7	61.5	69.9	75.3	76.7	70.9	63.6	55.9	50.1
	DBStd	12.10	6.63	6.94	7.28	6.53	5.32	3.98	2.98	3.01	4.17	5.26	6.13	6.63
	HDD50	522	143	142	114	27	0	0	0	0	0	1	16	79
	HDD65	2917	563	520	507	346	133	7	0	0	7	89	282	463
	CDD50	4191	45	43	75	138	357	598	784	827	628	423	191	82
	CDD65	1110	0	0	3	7	25	155	319	362	185	46	7	1
	CDH74	5687	0	3	37	76	53	414	1843	2542	602	88	25	4
	CDH80	598	0	0	10	20	9	18	170	322	37	10	2	0
(14) Wind	WSAvg	5.9	6.4	6.5	6.4	5.5	5.1	5.4	5.7	5.8	6.3	5.9	6.0	6.2
(15) Precipitation	PrecAvg	30.4	2.8	2.2	2.1	2.3	2.0	2.1	1.3	1.7	2.8	4.2	3.5	3.3
	PrecMax	35.9	6.2	5.6	4.2	4.3	3.8	5.8	4.6	3.4	5.9	11.5	8.0	6.9
	PrecMin	22.7	0.5	0.4	0.6	1.0	0.6	0.1	0.5	0.1	0.3	0.2	0.7	0.8
	PrecStd	3.8	1.5	1.3	1.1	0.9	0.8	1.2	0.9	1.0	1.5	2.6	1.9	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	73.2	78.6	82.6	80.0	81.8	84.6	85.9	83.8	80.9	77.3	73.1
		MCWB	51.0	53.0	54.4	61.2	65.3	72.5	75.9	77.3	72.9	64.0	56.0	53.8
	2%	DB	65.0	66.6	71.2	73.7	73.8	79.9	82.8	84.3	80.7	75.9	71.9	67.8
		MCWB	48.4	49.7	53.6	58.8	65.6	72.1	75.0	76.2	72.3	65.6	55.8	51.9
	5%	DB	60.9	62.6	64.5	66.7	71.3	78.1	82.1	82.8	78.9	73.5	67.8	63.1
		MCWB	47.1	49.8	51.6	56.5	64.6	71.1	74.5	75.1	71.2	65.6	57.1	51.2
	10%	DB	55.8	56.9	59.0	62.5	69.0	76.6	80.6	82.0	77.2	71.4	64.4	59.1
		MCWB	47.4	47.8	50.6	55.4	63.3	70.0	73.5	74.5	70.2	64.7	56.1	50.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	55.6	58.8	66.9	69.5	75.3	78.6	78.9	75.4	70.2	64.0	57.8
		MCDB	62.8	66.2	71.0	77.9	75.8	80.0	83.3	83.7	80.8	75.4	70.4	65.9
	2%	WB	51.0	52.3	54.7	59.8	67.0	73.4	76.9	77.5	73.6	68.3	60.6	55.3
		MCDB	58.6	63.1	66.6	69.4	71.6	78.3	81.9	82.8	79.2	73.3	65.5	61.3
	5%	WB	49.7	50.1	52.7	57.6	65.4	71.8	75.4	76.4	72.2	66.7	58.8	53.4
		MCDB	56.0	58.2	62.1	64.5	69.7	77.1	80.8	82.1	77.8	71.7	64.2	59.8
	10%	WB	48.3	48.6	51.0	55.7	63.8	70.2	74.2	75.3	70.8	65.1	57.1	51.7
		MCDB	54.8	55.4	58.0	61.9	68.1	75.4	79.9	81.1	76.5	70.4	62.6	57.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.2	10.7	10.4	8.9	8.9	8.4	8.2	9.2	9.2	10.3	9.8
		MCDBR	15.9	18.8	21.5	20.1	13.1	10.5	9.1	8.6	10.4	12.2	15.2	16.1
	5% WB	MCWBR	7.9	8.0	8.9	10.3	9.0	8.1	7.3	7.1	8.5	9.1	9.5	8.7
		MCDWR	12.7	15.9	18.2	16.7	11.0	9.1	8.4	7.8	9.8	10.8	11.9	12.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.371	0.411	0.472	0.524	0.490	0.460	0.477	0.497	0.439	0.441	0.383	0.361
		taud	2.240	2.122	1.988	1.904	2.068	2.219	2.208	2.137	2.283	2.184	2.302	2.303
	Edn at Noon	Edn at Noon	239	246	244	240	253	261	255	245	252	233	234	233
		Edh at Noon	31	40	51	60	52	44	45	47	38	37	29	27
(44) All-Sky Solar Radiation	RadAvg	582	854	1162	1463	1760	1997	1957	1724	1431	953	666	502	(44)
	RadStd	60	98	99	153	149	134	131	114	107	89	86	67	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.13	N/A	-3.26	N/A	N/A	N/A	-117	-257	+315	+169
(47) Regional (4 neighbors)	N/A	N/A	N/A	+1.17	N/A	N/A	N/A	-203	+167	+74

Nomenclature: See separate page